# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

## Department of Computer Science and Engineering

## ASSESSMENT TOOL 1 – C Code Output Prediction

| | | | |
|---|---|---|---|
| **Course Code:** | CSA0305 | **Course Title:** | Data Structures |
| **CO Assessed:** | CO1 – Imitation (BL3, BL4) | **Assessment:** | Assessment Tool 1 – C Code Output Prediction |
| **Weightage:** | 35% of CIA | **Total Marks:** | 100 |
| **CO1 Statement:** | *Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.* | | |
| **Student Name:** | T. Varun Tej | **Reg. No.:** | 19511217 |
| **Section / Batch:** | | **Date:** | 10/07/2026 |

### Code of Conduct

*I (T. Varun Tej/192511217)____________________________________(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: ______________________

### Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

| Section / Topic | Focus | Q Nos | Marks |
|---|---|---|---|
| **Linked data structure** | Basic data structure | 1 - 2 | 20 |
| **Primitive and Non-Primitive Data Structure - Linear and Non-Linear Data Structure** | Classifications of Linear data structure | 3 - 4 | 20 |
| **Search Data Structure** | Linear Search and Binary Search | 5 -6 | 20 |
| **Recursion, Performance analysis** | Recursion | 7 - 8 | 20 |
| **Array Data Structures** | Implementations Using Array data structure | 9 - 10 | 20 |
| | | **GRAND TOTAL:** | **100 Marks** |

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THOTA VARUN TEJ
192511217RA

QUESTIONS

- **Q1** C Code – Array Traversal 10 marks
- **Q2** C Code – Linear Search 10 marks
- **Q3** C Code – Binary Search Mid 10 marks
- **Q4** C Code – Pointer with Array 10 marks
- **Q5** C Code – Array Update 10 marks
- **Q6** C Code - Linked List Node 10 marks

Q1 **C Code – Array Traversal** 10 marks

```
#include <stdio.h>
int main() {
int arr[] = {5, 10, 15, 20};
int i, sum = 0;
for(i = 0; i < 4; i++)
sum += arr[i];
printf("%d", sum);
return 0;
}
```

Your Answer

50

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QUESTIONS

Q1
C Code – Array Traversal
10 marks

Q2
C Code – Linear Search
10 marks

Q3
C Code – Binary Search Mid
10 marks

Q4
C Code – Pointer with Array
10 marks

Q5
C Code – Array Update
10 marks

Q6
C Code - Linked List Node
10 marks

Q7
C Code - Time Complexity Loop

Q10 **C Code - Pointer Increment**

10 marks

```
#include<stdio.h>
int main(){
int a[]={5,10,15};
int *p=a;
p++;
printf("%d",*p);
}
```

**Your Answer**

10

Save Answer

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Course: **CSA0305RA**

**CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)** OPEN

DEBUGGING & OPTIMIZATION TASK

0/10 answered

Attempt →

**CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)** OPEN

DEBUGGING & OPTIMIZATION TASK

0/10 answered

Attempt →

**CO3_AT3_MCQ** OPEN

Multiple Choice Questions - Non Linear Data Structures

0/25 answered

Attempt →

**CO1_AT1_C CODE OUTPUT PREDICTION** OPEN ✓ Completed

C CODE OUTPUT PREDICTION

10/10 answered

Review →